Adar1 Biotech Strategy

Specialist Healthcare & Biotechnology Long/Short Strategy

Track Record: Mar 2019 - Apr 2026 (86 Months / 7.2 Years) | **Benchmark:** S&P 500 Total Return

ATHÉNÉE INVESTMENT

CAPITAL INTRODUCTION & THIRD-PARTY MARKETING

CAGR (Net)	Ann. Volatility	Sharpe Ratio	Max Drawdown	Corr. to S&P 500
34.92%	**15.60%**	**2.02**	**-10.63%**	**+0.34**
S&P 500: 15.99%	S&P 500: 16.75%	S&P 500: 0.97 | Sortino: 5.65	S&P 500: -23.87% | Calmar: 3.28	Alpha: 26.4% | Beta: 0.31

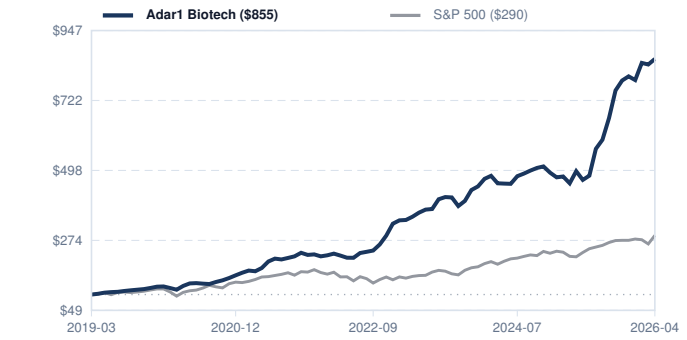
Performance, Risk & Hit Rate Indicators

Indicator	Fund	S&P 500
CAGR (Annualized)	**34.92%**	15.99%
Annualized Mean Return	31.48%	16.31%
Annualized Volatility	15.60%	16.75%
Proxy Sharpe Ratio ($r_f = 0\%$)	**2.02**	0.97
Annualized Sortino Ratio	5.65	1.60
Maximum Drawdown (MDD)	**-10.63%**	-23.87%
Calmar Ratio (CAGR / |MDD|)	3.28	0.67
Cumulative Total Return	755.41%	189.55%
Best / Worst Monthly	+17.90% / -6.70%	+12.82% / -12.35%
Positive Months (Hit Rate)	74.4% (64/86)	66.3% (57/86)
Crisis Hit Rate (Fund > 0 | SPX < 0)	**58.6% (17/29)**	–
Up-Mkt Loss (Fund < 0 | SPX > 0)	17.5% (10/57)	–

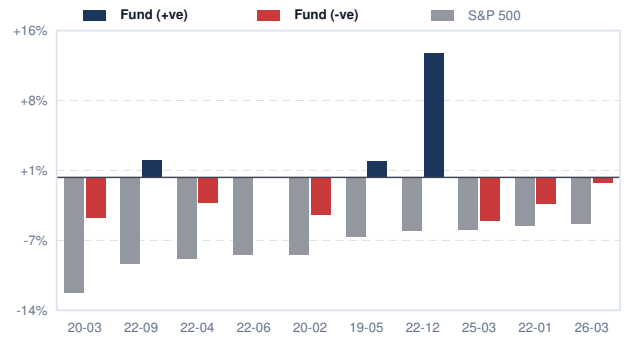
Alpha, Beta, Correlation & Tail Analysis (vs. S&P 500)

Quantitative Metric	Estimate / Diagnostic
Annualized Alpha (α) & t-stat	**26.38% (t=4.60)**
Market Beta (β_{SPX}) & t-stat	0.31 (t=3.26) ($R^2 = 11.3\%$)
Pearson Correlation (r) to S&P 500	**+0.34** (95% Blk CI: [0.11, 0.55])
Spearman Rank Correlation (ρ)	+0.34 (95% Blk CI: [0.09, 0.54])
Upside / Downside Capture Ratio	Up: **0.88** | Down: **-0.18**
Cond. Up-Market (SPX > 0)	$\alpha = 36.0\%,\ \beta = 0.14$
Cond. Down-Market (SPX < 0)	$\alpha = 27.6\%,\ \beta = 0.39$
Tail Corr. (SPX $< Q_1$, Worst 25%)	$r = 0.15,\ \rho = 0.16\ (n = 22)$
Tail Corr. (SPX $< D_1$, Worst 10%)	$r = 0.33,\ \rho = -0.07\ (n = 9)$
Worst 20% SPX Months Avg. Return	Fund: +0.38% vs SPX: -5.79%
Cross-Asset Beta (WTI / BTC)	WTI β: 0.03 | BTC β: 0.02
Ljung-Box Serial Correlation	$p_1 = 0.02,\ p_3 = 0.13$ (Serial Corr.)

Cumulative Growth of $100 (Fund vs. S&P 500)



Worst 20% S&P 500 Months (Top 10 Worst Months)



Historical Monthly & Yearly Net Return Matrix (%)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Yearly
2019	-	-	2.4%	3.4%	1.7%	0.9%	2.4%	2.0%	1.8%	1.5%	3.2%	3.3%	**25.0%**
2020	0.6%	-4.0%	-4.3%	10.3%	6.4%	0.6%	-1.1%	-0.9%	4.8%	3.8%	5.0%	5.7%	**29.1%**
2021	5.2%	4.1%	-1.2%	5.9%	11.3%	4.5%	-1.2%	2.3%	2.3%	5.1%	-2.9%	0.8%	**41.7%**
2022	-2.8%	1.5%	2.4%	-2.7%	-3.1%	-0.1%	7.1%	1.6%	1.8%	8.0%	10.9%	13.3%	**43.0%**
2023	3.2%	0.4%	3.1%	3.7%	2.7%	0.5%	8.4%	1.7%	-0.3%	-6.7%	4.3%	8.7%	**32.9%**
2024	2.7%	5.4%	2.1%	-5.0%	-0.2%	-0.2%	5.4%	1.7%	2.0%	1.7%	1.0%	-3.9%	**12.9%**
2025	-3.1%	0.6%	-4.6%	8.5%	-5.6%	2.9%	17.9%	5.1%	11.7%	13.3%	4.2%	1.7%	**62.9%**
2026	-1.5%	7.0%	-0.6%	2.1%	-	-	-	-	-	-	-	-	**7.0%**

Appendix, Quantitative Methodology & Risk Disclosures

Fund Reference: Adar1 Biotech Strategy (Mar 2019 - Apr 2026 (86 Months / 7.2 Years))

Appendix A: Extended Statistical, Bootstrap & Factor Diagnostics

Diagnostic Family	Methodology / Statistical Specification	Empirical Value
Pearson Correlation CI	95% Confidence Interval (IID Bootstrap vs. Moving Block $B = 3$, 1,000 iters)	Std: [0.14, 0.53] | Blk: [0.11, 0.55]
Spearman Rank Corr. CI	95% Confidence Interval (Rank Monotonic Co-Movement, IID vs. Block $B = 3$)	Std: [0.14, 0.53] | Blk: [0.09, 0.54]
Worst-Case Co-Dependence	97.5th Percentile Upper-Bound Block Bootstrap Correlation (PSD Projection)	$\rho_{\text{wc}} = 0.55$
Serial Correlation (Lag 1 & 3)	Ljung-Box Q-Statistic p-values testing for return smoothing / illiquidity bias	$p_1 = 0.02,\ p_3 = 0.13$ (Serial Corr.)
Conditional Up-Market OLS	Annualized Alpha & Beta estimated during positive S&P 500 months ($n = 57$)	$\alpha = 36.0\%,\ \beta = 0.14\ (t_\alpha = 2.97)$
Conditional Down-Market OLS	Annualized Alpha & Beta estimated during negative S&P 500 months ($n = 29$)	$\alpha = 27.6\%,\ \beta = 0.39\ (t_\alpha = 1.73)$
Commodity Factor (WTI)	OLS Regression against Front-Month WTI Crude Oil Monthly Returns	$\beta = 0.03\ (t = 0.98),\ \alpha = 30.8\%$
Digital Asset Factor (BTC)	OLS Regression against Bitcoin (BTC-USD) Monthly Returns	$\beta = 0.02\ (t = 0.80),\ \alpha = 30.2\%$

Appendix B: Quantitative Framework & Indicator Definitions

- **CAGR & Annualized Volatility:** Compound Annual Growth Rate is computed geometrically as CAGR $= \left(\prod_{t=1}^{T}(1 + r_t)\right)^{12/T} - 1$. Annualized Volatility is sample standard deviation scaled by $\sqrt{12}$: $\sigma_{\text{ann}} = \hat{\sigma}_m\sqrt{12}$. Proxy Sharpe Ratio assumes $r_f = 0\%$ ($\bar{r}_{\text{ann}}/\sigma_{\text{ann}}$).
- **Maximum Drawdown (MDD) & Calmar Ratio:** Maximum peak-to-trough decline of cumulative compounded wealth $W_t = \prod_{s\le t}(1 + r_s)$, defined as MDD $= \min_t\left((W_t - \max_{s\le t} W_s)/\max_{s\le t} W_s\right)$. Calmar Ratio is CAGR/|MDD|.
- **OLS Alpha (α), Beta (β) & Capture Ratios:** Estimated via Ordinary Least Squares $r_{\text{fund},t} = \alpha_m + \beta r_{\text{SPX},t} + \varepsilon_t$, where $\alpha_{\text{ann}} = 12 \times \alpha_m$. Upside (Downside) Capture measures the ratio of the fund's mean return to the S&P 500's mean return during positive (negative) S&P 500 months.
- **Tail Correlations & Moving Block Bootstrap:** Tail correlations evaluate linear (r) and rank (ρ) co-movement during S&P 500 drawdowns below the 25th percentile (Q_1) and 10th percentile (D_1). Moving Block Bootstrap ($B = 3$ months) preserves local serial dependencies when constructing 95% confidence intervals.

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